



ST6002CEM-Mobile Application Development

Project Proposal

On

Home Tutor App



Submitted by: Shreedhar Chaudhary

Coventry ID: 230070

Batch: 35C

Module Leader: Kiran Rana

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Project Introduction

The Home Tutor App is an educational service that runs on mobile devices and connects students with professional tutors for individual academic help. The app's goal is to ease the process of learning by providing personalized sessions, flexibility in scheduling, and materials chosen according to each student's needs. Besides, the app applies the modern UI/UX style to create a user-friendly interface, and the MVVM architectural pattern for structured, scalable development.

Justification of the Project

Background of the Project

The digital learning trend has caused a lot of students to have a hard time due to lack of proper online tutors, and these are not easy to find for every need and pace of learning. Fixed timing and limited students' choice are often the main drawbacks of traditional tutoring that require students' physical presence. Nowadays, parents and learners prefer online platforms that provide convenience, safety, and personalized learning experiences. The main goal of this project is to eliminate that barrier and to provide a trustworthy online space where students can easily find tutors, schedule sessions, and learn effectively.

Problem Statement

Students often encounter problems like having few professional tutors available to them, not having individualized study plans, and the long process of trying to find one. Tutors in contrast have a hard time contacting the right students. Current applications are either too complex, too pricey, or do not provide basic communication facilities. The Home Tutor App intends to eliminate all these problems and provide a straightforward, user-friendly, and trustworthy platform for both students and teachers.

Description of the Project

Features of the Project

- **User Authentication:** Secure login and signup through email, phone number, or third-party platforms.
- **Tutor Discovery:** Students can browse tutors by subject, expertise, ratings, and price.
- **Session Booking:** Easy scheduling with real-time availability.
- **Payment System:** Fast and secure payments through integrated gateways.
- **Personal Dashboard:** Students can track courses, progress, and upcoming sessions.
- **Notifications:** Alerts for session reminders, tutor updates, and new course announcements.
- **Study Plan Creation:** Students can build customized study plans according to their learning goals.
- **Interactive UI:** Smooth navigation with modern design and minimalistic screens for easy use.

Aims and Objective

The main goal is to develop an efficient and friendly mobile application that will easily link students to home tutors. Several objectives are set: Customized learning experience for the students. An opportunity for the tutors to present their talents and to work with flexible payments. Quick, safe, and convenient booking and payment processes. Overcoming communication barriers between tutors and students with an easy-to-use interface. Transforming the entire educational process by offering structured study plans and monitoring progress.

MVVM Design Pattern

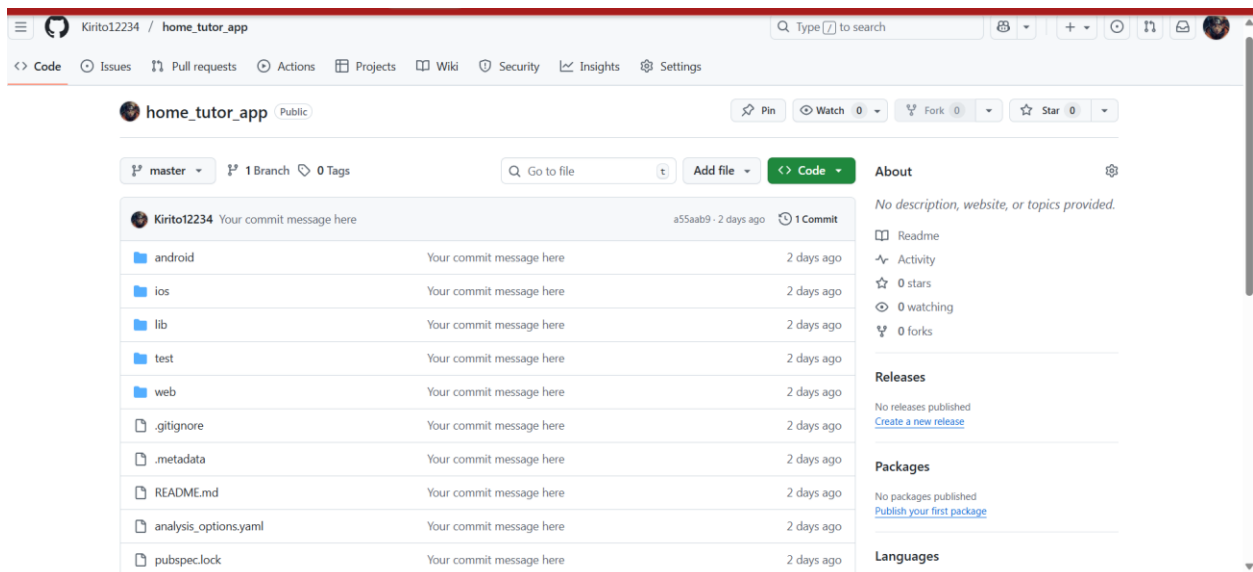
The application uses the Model–View–View Model (MVVM) architecture which guarantees a clean and maintainable codebase. Model: Manages data logic including user information, tutor details, booking history, and interactions with APIs. View: Comprises the UI elements that are developed with modern and simple layouts. The business logic is completely excluded while each screen is updated with the data in real-time. View Model: Serves as the link between the Model and the View. It performs data processing, manages user interactions, and uses observable data streams to communicate changes in the user interface. This arrangement minimizes coupling, increases performance, and makes debugging and future growth easier.

Configuration Management

GitHub Link

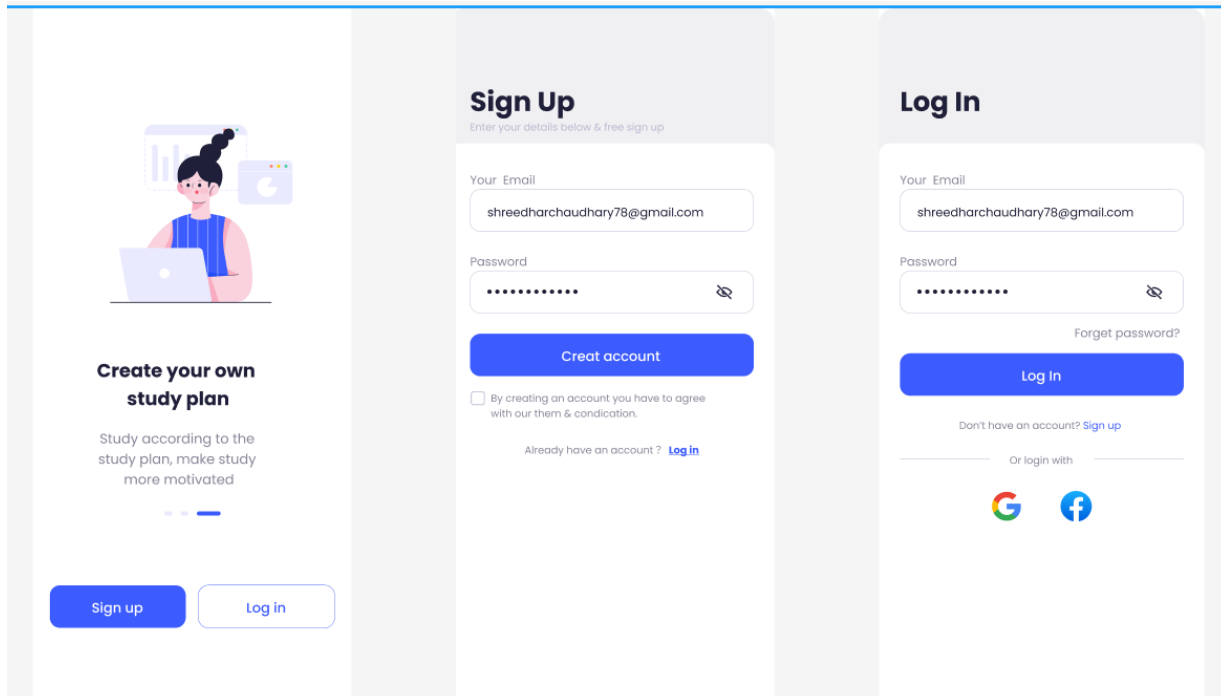
https://github.com/Kirito12234/home_tutor_app.git

GitHub Repository Screenshot

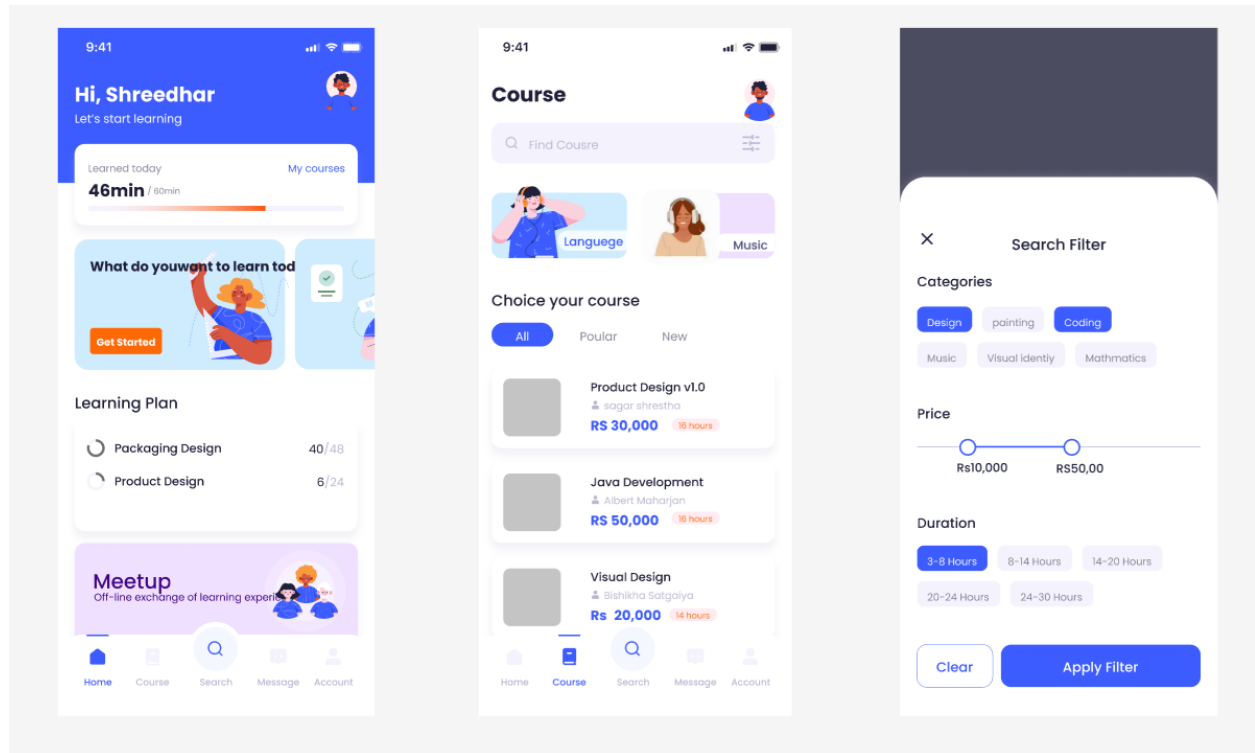


Prototyping

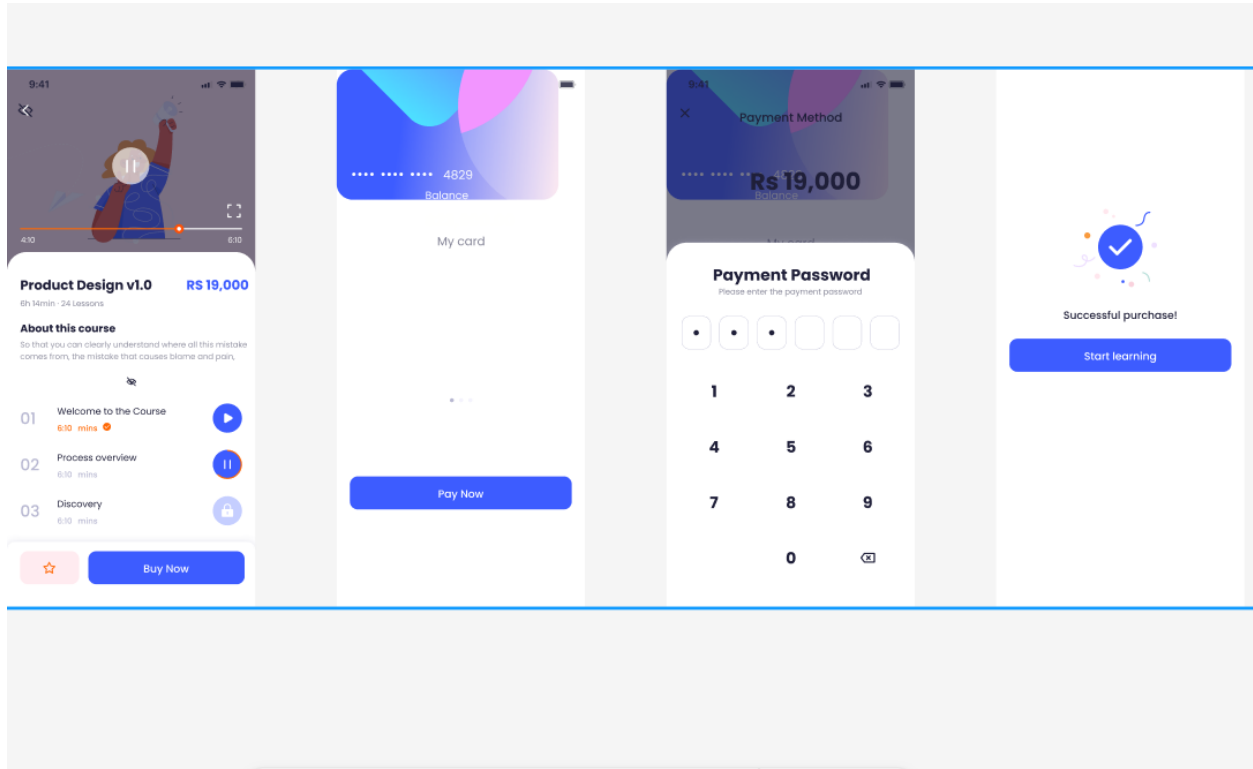
The design phase involved creating interactive mockups using Figma. These prototypes included: Onboarding screens explaining the app's purpose.



Tutor browsing pages with subject filters, pricing, and ratings.



Payment, Progress Dashboards, and Notifications



The prototypes helped visualize the user journey, ensuring smooth navigation and a consistent visual identity throughout the app.

Conclusion

The Home Tutor Application brings an advanced answer to the difficulties that are inherent in the old-fashioned tutoring methods. By means of user interface design that is easy to understand, features that are organized, and an MVVM architecture that grows in capacity, the app has been able to eliminate all the barriers that exist between the students and the tutors. It guarantees that learning is not restricted to certain hours, that education is always within reach, and that teachers have a good place to exhibit their knowledge. The endeavor is not only a show-off of the technical skills it has developed but also a solution that caters to the educational needs in a very practical and relevant way.

References

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